

2023 Ch H1 Q17

Section: Researching Chemistry

Topic: Experimental Procedures

Question Summary

The apparatus was used to measure the enthalpy of combustion of ethanol. Which of the following would not improve the accuracy of the result?

- A: Using a draught shield
- B: Moving the thermometer
- C: Using a glass beaker instead of a copper can
- D: Stirring the water

Worked Solution

- Using a draught shield reduces heat loss to the surroundings → improves accuracy.
- Moving the thermometer (ensuring it is properly positioned) gives more reliable temperature readings → improves accuracy.
- Stirring the water distributes heat evenly → improves accuracy.

- Replacing the copper can with a glass beaker would decrease accuracy, because glass is a poorer conductor of heat than copper.

Final Answer

C — Using a glass beaker instead of a copper can

Revision Tips

- Copper is used because it is a good conductor of heat.
- Always think: does the change reduce heat loss or improve heat transfer?
- In calorimetry experiments, the main errors come from heat lost to the surroundings.